

Zero-Shot Vision-Language Models for Classroom Engagement Recognition: A Benchmark Study of Prompt Sensitivity and Cross-Dataset Generalization

Aman Goyal*
Carnegie Mellon University

Kshama Nitin Shah*
University of Michigan, Ann Arbor

Kemmannu Vineet Venkatesh Rao
Magna International

Abstract

Automated classroom engagement recognition holds substantial promise for scalable learning analytics, yet the suitability of modern Vision-Language Models (VLMs) for this task under zero-shot conditions remains largely unexplored. We present a systematic benchmark that evaluates five widely-used VLMs—CLIP, BLIP-VQA, GPT-4o, LLaVA-1.5-7B, Qwen2.5VL-7B-Instruct—across two complementary educational datasets: DAiSEE, an individual-student video dataset (300 sampled test clips), and the Student Classroom Behaviour dataset (SCB, 1,168 scene-level images). Each model is probed with three prompt variants spanning minimal, rubric-anchored, and chain-of-thought designs. Our experiments reveal three primary failure modes of zero-shot VLMs for engagement recognition: (1) near-random performance on individual students, with Cohen’s κ never exceeding 0.10 on DAiSEE; (2) severe class collapse, where models assign 85–100% of predictions to a single engagement level regardless of visual content; and (3) extreme prompt sensitivity, with accuracy swings of up to 32 percentage points on identical images depending solely on prompt phrasing. Remarkably, scene-level classification on SCB is substantially more tractable: CLIP and GPT-4o achieve $\kappa \approx 0.60$ when prompted with behaviorally-grounded rubrics. We also document a practical barrier for deployment—GPT-4o’s safety filters reject 98% of chain-of-thought requests involving individual student faces. Our findings provide a calibrated baseline and surface critical design considerations for the use of VLMs in educational observation systems.

1. Introduction

Understanding student engagement is central to effective teaching. Disengagement—whether manifested as wandering attention, passive participation, or behavioural withdrawal—predicts lower learning outcomes and higher attrition [3]. Traditionally, measuring engagement at scale requires trained human observers [14], a resource-intensive process that is difficult to replicate in large lecture halls or remote learning environments. The rapid proliferation of cameras in educational settings, coupled with powerful visual recognition models, has therefore sparked interest in automated engagement sensing [4, 9, 17, 20].

Contemporary Vision-Language Models (VLMs) represent a potential leap forward for this problem. Unlike supervised classifiers that require large labelled corpora specific to each school or cultural context, models such as CLIP [15], BLIP [8], GPT-4o [13], LLaVA [10] and Qwen [2] can be queried zero-shot via natural language, with no task-specific fine-tuning. Their ability to reason from open prompts makes them attractive as *plug-and-play* educational observers. Yet, to our knowledge, no prior work has systematically benchmarked VLMs for classroom engagement recognition under controlled, zero-shot conditions.

Our contributions. This paper presents the following:

- A zero-shot benchmark of five VLMs on two complementary datasets: the individual-student DAiSEE dataset [4] and the scene-level Student Classroom Behaviour (SCB) dataset [18], using three systematically-designed prompt variants (minimal, rubric, chain-of-thought).
- A rigorous quantitative analysis documenting class collapse, near-random agreement on individual-level data ($\kappa \leq 0.10$), and substantial prompt sensitivity (accuracy variance up to 32 pp).
- The finding that scene-level engagement classification is significantly more tractable for VLMs, with CLIP achieving $\kappa = 0.60$ and GPT-4o achieving $\kappa = 0.58$ on SCB.

*Equal contribution.

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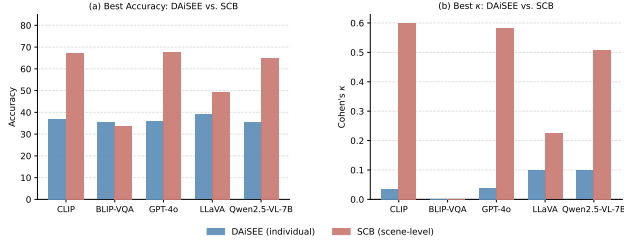


Figure 1. Best zero-shot accuracy and Cohen’s κ per model on DAiSEE (individual student, video frames) versus SCB (classroom scene images). Scene-level classification is substantially more tractable, with CLIP and GPT-4o achieving $\kappa \approx 0.60$ on SCB while remaining near-random on DAiSEE.

- Documentation of GPT-4o’s safety-guardrail refusals blocking chain-of-thought engagement analysis for individual student faces, with practical implications for educational AI deployment.
- A self-consistency analysis showing that generative VLMs (GPT-4o, LLaVA) change their predictions for 27–32% of images across repeated runs at temperature $T = 0.7$, raising reliability concerns for classroom monitoring applications.

Our results paint a nuanced picture: while VLMs are not yet reliable individual engagement observers, they demonstrate meaningful capability at the scene level. This gap between individual and scene-level performance has direct implications for system design—aggregating information spatially before querying a VLM may be a more productive paradigm than per-student analysis.

2. Related Work

Engagement detection in educational settings. Early work by Whitehill *et al.* [20] demonstrated that facial features encode cognitive states relevant to learning. The release of DAiSEE [4] established the first large publicly-available benchmark for four-level engagement classification from student video. Subsequent supervised methods leveraged spatiotemporal features from video [9], recurrent networks over facial action units [17], and attention-based architectures [5]. More recently, spatiotemporal and attention-based models trained on DAiSEE have reported accuracy of 55–65% [5, 9], underscoring the difficulty even for task-specific supervised models. Classroom-level engagement has received comparatively less attention, though several works address student behaviour recognition from overhead cameras [6, 18].

Zero-shot visual recognition with VLMs. CLIP [15] pioneered zero-shot image classification via image-text cosine similarity, demonstrating competitive performance on many standard benchmarks. BLIP [8] introduced bootstrapped

image-text alignment that enabled visual question answering without task-specific training. Instruction-tuned generative models such as LLaVA [10] and GPT-4V/4o [13] further extend this to open-ended visual reasoning. A growing body of work has applied these models to medical imaging [22], document understanding, and remote sensing, but their application to affect recognition and educational observation has not been systematically studied.

Prompt design and sensitivity. The sensitivity of VLMs to prompt phrasing has been studied in the context of natural language processing [11, 23], but visual prompting for classification remains less understood. Studies on zero-shot classification [12] demonstrate that performance can vary by 10–30 percentage points across prompt phrasings—a phenomenon we observe and quantify here.

3. Methodology

3.1. Datasets

DAiSEE. The Dataset for Affective States in E-Environments (DAiSEE) [4] contains 9,068 ten-second video clips from 112 subjects recorded via webcam during e-learning sessions. Each clip is annotated on a four-point Likert scale for engagement (0 = not engaged, 3 = highly engaged) along with boredom, confusion, and frustration. We focus exclusively on engagement. The official test split comprises 1,784 clips across 21 subjects; the label distribution is heavily skewed, with only 4 samples at level 0 and approximately 850 each at levels 2 and 3.

We evaluate on a stratified sample of $N = 300$ test clips, retaining all 4 level-0 clips, all 84 level-1 clips, and randomly sampling 106 clips each from levels 2 and 3. From each 10-second clip, we extract a single representative frame at $t = 5$ s (the temporal midpoint) using `ffmpeg`. This setup tests whether static visual cues visible in a single webcam frame are sufficient for zero-shot engagement inference.

Student Classroom Behaviour (SCB). The SCB dataset [18] contains classroom scene images with per-student bounding box annotations across six behaviour categories: *reading*, *writing*, *raising hand*, *using phone*, *bowing head*, and *leaning over table*. We use a combined split of 1,168 validation images drawn from both the original 5K corpus and an additional university subset.

Following the aggregation strategy proposed in the dataset documentation, we derive a scene-level engagement label from the per-student annotations. Each behaviour is mapped to an engagement score (0–3): disruptive behaviours (*phone*, *bowing head*) score 0; passive disengagement (*leaning over*) scores 1; productive behaviours (*reading*, *writing*) score 2; and active participation (*raising hand*) scores 3.

The scene-level label is the mean across all annotated students, discretised via uniform thresholds ($[0, 0.75) \rightarrow 0$; $[0.75, 1.75) \rightarrow 1$; $[1.75, 2.75) \rightarrow 2$; $[2.75, 3] \rightarrow 3$). The resulting label distribution is highly imbalanced: 54.5% of images fall at level 3, 33.6% at level 2, 10.1% at level 1, and 1.9% at level 0.

Task formulation. We note that our evaluation of SCB differs fundamentally from prior work on this dataset. Existing methods [16, 18] treat SCB as a *per-student object detection* task: they first localise individual students via bounding-box detectors (e.g., YOLOv5 [18]) and then classify each detected student’s behaviour independently. In contrast, we formulate engagement recognition as *scene-level image classification*, where the model receives the full classroom image and produces a single holistic engagement score aggregated across all visible students. This reformulation is deliberate: zero-shot VLMs are not designed for spatial object detection, and our results (Sec. 4.2) demonstrate that scene-level cues—spatial layout, group dynamics, and overtly visible behaviours—are precisely the signals that make zero-shot engagement recognition tractable ($\kappa \approx 0.60$). Because these are fundamentally different task formulations operating at different output granularities, direct numerical comparison between our scene-level accuracy and prior per-student detection metrics is not meaningful.

3.2. Models

We evaluate four models at substantially different points on the capability-cost spectrum:

CLIP (ViT-B/32) [15] is a discriminative contrastive model. Classification proceeds by computing cosine similarity between the image embedding and embeddings of textual descriptions of each engagement level; the class with the highest similarity is the prediction. Inference is fully deterministic.

BLIP-VQA (blip-vqa-base) [8] is a discriminative VQA model. We pose a natural-language question about student engagement; the model generates a short free-text answer (e.g., “not very”, “yes”) that is mapped to a level via a hand-crafted lookup table. Beam decoding with $k = 3$ ensures deterministic output.

GPT-4o [13] is a proprietary closed-source multimodal model accessed via the OpenAI API. Images are submitted as base64-encoded JPEG at “low” detail resolution. Temperature is 0.0 for run 1 and 0.7 for subsequent consistency runs.

LLaVA-1.5-7B [10] is an open-source instruction-tuned model served locally through Ollama at 4-bit quantisation (reducing memory from ~ 14 GB to ~ 4 GB) on a local machine with 16 GB RAM. We use the same prompts as GPT-4o to enable direct comparison. Temperature follows the same protocol as GPT-4o.

Qwen2.5-VL-7B-Instruct [2] is an open-source vision-language model from Alibaba’s Qwen team, loaded in bfloat16 precision via `Qwen2.5-VLForConditionalGeneration` from the Hugging Face Transformers library [21]. We use the same three prompt variants as GPT-4o and LLaVA to enable direct comparison. Temperature follows the same protocol: $T = 0.0$ for run 1 (deterministic) and $T = 0.7$ for self-consistency runs 2 and 3.

3.3. Prompt Variants

Three prompts are applied to each model to quantify sensitivity:

P1 (Minimal) provides only a four-label numeric scale with no contextual guidance. It establishes a lower bound on what each model can infer from minimal language grounding.

P2 (Rubric-Anchored) augments P1 with explicit behavioural anchors for each level, e.g., “*looking away, yawning*” for level 0 and “*leaning forward, showing curiosity*” for level 3. This mirrors how human annotators are typically trained.

P3 (Chain-of-Thought) instructs the model to first describe the student’s or class’s observed facial expression and body posture before committing to a rating. This structured reasoning format has shown benefits in language-only tasks [19] and is evaluated here in the visual domain.

For CLIP, the prompts are class-specific text templates. For BLIP-VQA, the prompts are question strings. For GPT-4o and LLaVA, identical prompt text is used to enable direct cross-model comparison.

3.4. Evaluation Metrics

All experiments are evaluated with four metrics: (i) **Accuracy** (exact-match 4-class); (ii) **F1 Macro**, the unweighted mean of per-class F1 scores, which penalises class collapse; (iii) **Cohen’s κ (quadratic weighted)**, which accounts for ordinal agreement structure and penalises predictions far from ground truth more severely; and (iv) **MSE** (mean squared error), measuring average ordinal distance. For self-consistency, we report full 3-run agreement rate and pairwise agreement rate.

4. Experimental Results

4.1. DAiSEE: Individual-Student Engagement

Table 1 shows the best-performing configuration for each model alongside published supervised baselines. The gap is striking: even the strongest VLM achieves only 39.0% accuracy and $\kappa = 0.10$ (LLaVA with chain-of-thought), while supervised convolutional+temporal models reach 58.8% [9] and context-aware transformers achieve $\kappa > 0.35$ [5].

Table 1. Best classification results on DAiSEE ($N = 300$ test clips). Supervised baselines are drawn from published work on the full test set.

Best VLM results are highlighted in **bold**.

Model	Acc. (%)	F1 Macro	κ	MSE
<i>Supervised baselines (literature)</i>				
ResNet+TCN [9]	58.8	—	—	0.18
Huang <i>et al.</i> [5]	62.3	—	0.37	—
CavT [1]	—	—	—	0.038
<i>Zero-shot VLMs (this work)</i>				
CLIP (P1)	37.0	0.224	0.036	1.00
BLIP-VQA (P3)	35.3	0.131	0.000	0.69
GPT-4o (P3) [†]	36.0	0.153	0.039	0.69
LLaVA-1.5 (P3)	39.0	0.210	0.101	0.72
Qwen2.5-VL-7B-Instruct (P3)	35.3	0.136	−0.041	0.737

[†]98% of GPT-4o P3 responses were safety refusals; results reflect fallback-to-level-2 parsing.

Table 2. Full prompt-sensitivity results on DAiSEE. Pred. Dist. shows the fraction of predictions in each class (GT: 1%–28%–35%–35%).

Model	Prompt	Acc.	F1	κ	MSE
CLIP	P1	37.0	0.224	0.036	1.00
	P2	31.3	0.184	0.068	1.39
	P3	35.0	0.231	−0.042	1.21
BLIP-VQA	P1	25.3	0.116	−0.018	2.39
	P2	32.7	0.153	0.058	0.94
	P3	35.3	0.131	0.000	0.69
GPT-4o	P1	27.7	0.144	0.067	1.67
	P2	32.3	0.192	0.075	1.51
	P3 [†]	36.0	0.153	0.039	0.69
LLaVA-1.5	P1	5.3	0.047	0.037	4.07
	P2	15.7	0.097	0.045	2.84
	P3	39.0	0.210	0.101	0.72
Qwen2.5-VL-7B-Instruct	P1	30.0	0.25	0.098	1.34
	P2	30.0	0.257	0.097	1.30
	P3	35.3	0.136	−0.041	0.737

[†]Safety refusal rate $\approx$ 98%; level-2 fallback inflates accuracy.

Table 2 presents the full prompt-by-prompt breakdown. Several patterns stand out. First, no single model-prompt combination surpasses $\kappa = 0.11$, confirming near-random ordinal agreement across the board. Second, all models show accuracy near or below the majority-class baseline of 35.3% (level 2), suggesting the higher accuracies reflect correct predictions of the modal class rather than genuine four-class discrimination.

Table 3. Results on SCB ($N = 1,168$ validation images). All numbers are the best single prompt per model.

Model	Best Prompt	Acc. (%)	κ	MSE
CLIP (ViT-B/32)	P3	67.3	0.599	0.586
BLIP-VQA	P1/P2/P3*	33.6	0.000	0.721
GPT-4o	P2	67.9	0.582	0.372
LLaVA-1.5	P2	49.2	0.224	2.666
Qwen2.5-VL-7B-Instruct	P2	64.9	0.508	0.408

*BLIP-VQA produces identical predictions across all three prompts on SCB (complete zero sensitivity; see Sec. 4.3).

Table 4. Full prompt-sensitivity results on SCB.

Model	Prompt	Acc.	F1	κ	MSE
CLIP	P1	14.4	0.097	0.100	1.926
	P2	24.7	0.172	0.263	2.564
	P3	67.3	0.472	0.599	0.586
BLIP-VQA	P1	33.6	0.126	0.000	0.721
	P2	33.6	0.126	0.000	0.721
	P3	33.6	0.126	0.000	0.721
GPT-4o	P1	39.1	0.211	0.154	0.688
	P2	67.9	0.400	0.582	0.372
	P3	51.8	0.273	0.260	0.603
LLaVA-1.5	P1	7.4	0.067	0.027	5.343
	P2	49.2	0.353	0.224	2.666
	P3	33.9	0.131	−0.027	0.762
Qwen2.5-VL-7B-Instruct	P1	50.5	0.283	0.298	0.586
	P2	64.9	0.365	0.508	0.408
	P3	33.6	0.126	0.00	0.721

4.2. SCB: Scene-Level Classroom Engagement

Table 3 shows results on the SCB dataset. Performance improves dramatically compared with DAiSEE. CLIP with chain-of-thought templates reaches $\kappa = 0.60$ and GPT-4o with the rubric prompt reaches $\kappa = 0.58$ —placing both in the “moderate” agreement range [7]. These numbers approach the performance of supervised methods in comparable scene-understanding tasks.

The full SCB prompt breakdown (Table 4) reveals interesting cross-prompt dynamics. CLIP shows a dramatic positive response to the emotional/behavioural template in P3 (from $\kappa = 0.10$ at P1 to $\kappa = 0.60$ at P3). GPT-4o peaks with the rubric prompt (P2) but drops 32 pp in accuracy under chain-of-thought (P3). LLaVA performs best under P2 on SCB, unlike on DAiSEE where P3 is superior.

4.3. Analysis of Failure Modes

Class collapse. Figure 3 illustrates the predicted class distributions. BLIP-VQA with P3 assigns *all* 300 DAiSEE samples to level 2, achieving 35.3% accuracy solely by matching the plurality class. GPT-4o concentrates 82–95% of predic-

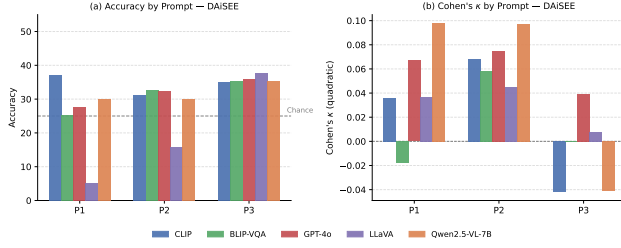


Figure 2. Accuracy and Cohen’s κ across the three prompt variants on DAiSEE. LLaVA’s accuracy spans from 5.3% (P1) to 39.0% (P3)—a 33.7 pp swing on identical images. Dashed line in (a) marks the four-class random-chance baseline (25%).

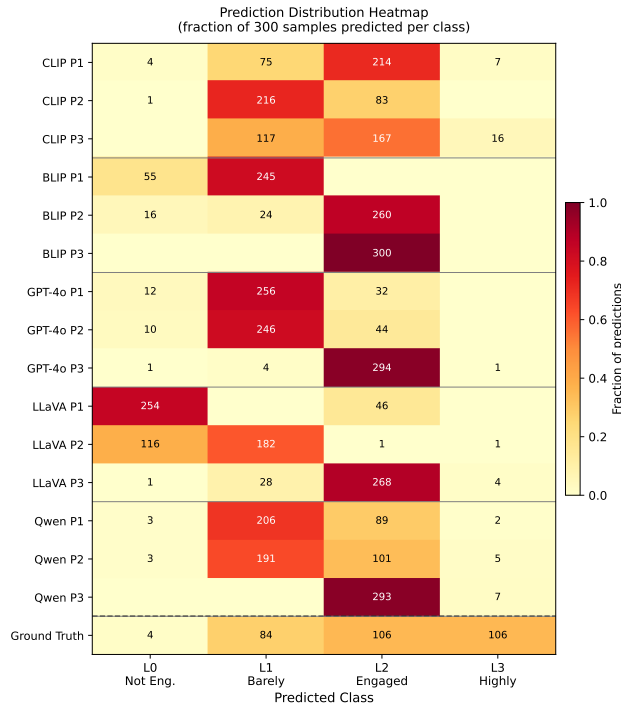


Figure 3. Fraction of the 300 DAiSEE samples predicted at each engagement level, for every model-prompt combination and the ground truth distribution. Warm colours indicate high concentration. Each model collapses to 1–2 dominant classes, confirming class collapse rather than genuine multi-class discrimination.

tions on level 1 across P1 and P2, suggesting a systematic low-engagement bias. No model ever assigns more than 5% of samples to level 3 (highly engaged) despite it comprising 35% of the ground truth. This asymmetric collapse is not random noise; it reflects each model’s learned prior over language space that “engaged” or “barely engaged” are more plausible descriptions of a sitting student than “highly engaged.”

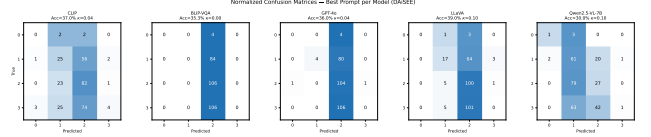


Figure 4. Normalised confusion matrices (rows = true label, cols = predicted label) for the best prompt per model on DAiSEE. Cell values show raw counts. All models achieve near-zero recall on levels 0 and 3, confirming class collapse.

Table 5. Prompt sensitivity (max – min across P1/P2/P3) for each model and dataset. Higher values indicate greater dependence on prompt phrasing.

Model	DAiSEE			SCB		
	ΔAcc	$\Delta\kappa$	ΔMSE	ΔAcc	$\Delta\kappa$	ΔMSE
CLIP	5.7	0.110	0.389	52.9	0.499	1.978
BLIP-VQA	10.0	0.076	1.703	0.0	0.000	0.000
GPT-4o	8.3	0.060	0.987	28.8	0.428	0.316
LLaVA	32.3	0.037	3.313	41.8	0.251	4.581

Confusion matrix analysis. Figure 4 shows normalised confusion matrices for the best configuration of each model. All models achieve reasonable recall on level 2 (the largest class) but near-zero recall on levels 0 and 3. LLaVA shows the broadest distribution, occasionally predicting level 1, but still fails on level 3 entirely. The diagonal patterns confirm that apparent accuracy gains are driven by level-2 majority-class capture rather than calibrated multi-level discrimination.

Prompt sensitivity. Table 5 quantifies the range of each model’s performance across the three prompts. LLaVA exhibits the highest sensitivity: a 32.3 pp accuracy swing on DAiSEE and 3.31 MSE variation. On SCB the effect is even more pronounced for CLIP (53 pp swing from P1 to P3). BLIP-VQA is the most stable model—but for the wrong reason: it is insensitive because it *always* predicts the same label regardless of prompt or image.

Self-consistency. We ran the best prompt three times with temperature $T = 0.7$ for GPT-4o (P2) and LLaVA (P3). CLIP and BLIP-VQA are deterministic and trivially achieve 100% agreement. GPT-4o achieves full 3-run agreement on only 68.0% of images (pairwise: 79.0%), while LLaVA achieves 73.3% (pairwise: 82.2%). Kappa across LLaVA runs spans from 0.008 to 0.101, meaning a single run’s reported kappa is unreliable as an estimate of model quality. GPT-4o additionally triggered intermittent safety refusals in run 3 that did not occur in runs 1 or 2, further degrading consistency.

GPT-4o safety guardrails. Chain-of-thought prompting (P3) asks the model to describe “the student’s facial expression,” which GPT-4o interprets as facial recognition of a real person. This triggers its safety filters in $\approx 98\%$ of DAiSEE images: the model responds with “I’m unable to assist with requests involving identifying or describing individuals from their photographs.” Our parsing pipeline falls back to level 2 for all refusals, artificially inflating P3 accuracy to 36% while providing zero true information. This is a practical barrier for any system deploying GPT-4o as an individual student observer. No such refusals occur with the scene-level SCB images.

4.4. Feature Space Analysis

To understand the failure modes mechanistically, we analyse CLIP’s internal representations directly, extracting ViT-B/32 image embeddings for all 300 DAiSEE frames and examining both feature geometry and text-image similarity structure.

Feature geometry. Figure 5 shows a t-SNE projection of image embeddings coloured by ground-truth engagement label. Frames cluster by *student identity and visual appearance* rather than engagement level—all four engagement classes intermix within every cluster. CLIP organises the representation space around “who this person is” (lighting, face, background) rather than “how engaged they are.” This provides a direct geometric explanation for why zero-shot classification fails: the engagement signal does not correspond to a separable direction in CLIP’s feature space.

Text-image similarity collapse. Figure 6 shows cosine similarities between image embeddings and each of the four class text templates, stratified by true engagement class, across all three prompt variants. Two observations are consistent across all prompts: (1) the correct text template (shaded column) does not yield the highest similarity for any true class—the diagonal is never dominant; (2) the total similarity range spans only ≈ 0.07 , meaning all four text embeddings are nearly equidistant from every image. In this regime, argmax predictions are determined by which text template happens to land marginally closer in embedding space, irrespective of image content. Concretely, for P1 the L2 template wins globally, collapsing all predictions to “Engaged”; for P2, L1 dominates; for P3, L1 and L2 differ by < 0.003 , producing high run-to-run variance. This directly explains both the class collapse and the extreme prompt sensitivity reported in Tables 2 and 5.

5. Discussion

Why does scene-level outperform individual-level? The performance gap between DAiSEE ($\kappa \leq 0.10$) and SCB (κ

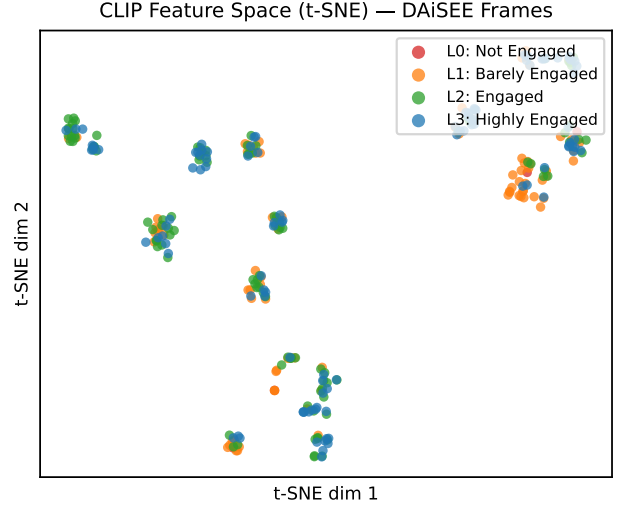


Figure 5. t-SNE projection of CLIP ViT-B/32 image embeddings for all 300 DAiSEE frames, coloured by ground-truth engagement label. The four engagement classes intermix within every cluster, confirming that CLIP organises frames by subject identity rather than engagement state.

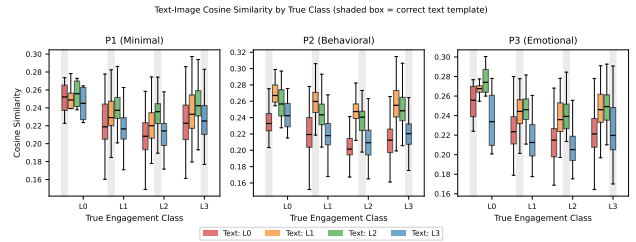


Figure 6. Text-image cosine similarity distributions across all three prompt variants, stratified by true engagement class. Each group of four boxes shows the similarity to text templates L0–L3; the shaded box marks the *correct* template. Distributions overlap heavily across all prompts and the correct template never achieves the highest median—confirming that CLIP has no discriminative signal for individual engagement level.

up to 0.60) is not simply a function of dataset difficulty. We hypothesise three contributing factors. First, *behavioural salience*: SCB labels derive from overt physical actions (raised hand, phone use) that are visually unambiguous to a language-grounded model, whereas DAiSEE labels reflect internal cognitive states that manifest only subtly in facial microexpressions. Second, *context richness*: a classroom scene provides spatial layout, group dynamics, and object context (e.g., desks, boards) that help ground engagement inference, while a webcam close-up of a single face offers minimal contextual anchors. Third, *label construction*: the SCB aggregation scheme maps explicit physical categories to ordinal scores, creating a more objective signal than sub-

jective per-clip engagement ratings.

Implications for system design. Our findings suggest that deploying VLMs as individual student observers in their current zero-shot form is premature. A more productive paradigm may be *scene-first analysis*: aggregate visual information at the classroom level, then query a VLM. This approach aligns better with how teachers actually monitor classrooms—by scanning the room rather than studying individual faces. For privacy reasons, scene-level analysis is also preferable as it avoids close facial analysis of minors.

Prompt engineering as unreliable compensation. The wide prompt sensitivity we observe means that practitioners who tune prompts on a small validation set risk over-fitting to that distribution. A prompt that yields 67.9% accuracy for GPT-4o on SCB could easily degrade by 28 pp on a different class distribution. Reliable deployment requires either task-specific fine-tuning or principled prompt selection with held-out validation.

Open vs. closed source. LLaVA-1.5-7B (4-bit quantised, running locally) matches or outperforms GPT-4o on DAiSEE, and achieves competitive performance on SCB at zero API cost and without data leaving the institution. This is encouraging for privacy-sensitive educational deployments, though the comparison is not perfectly controlled given the quantisation difference.

Limitations. This study is limited to zero-shot evaluation. Fine-tuning VLMs on even modest task-specific data would likely close the gap substantially. The single-frame assumption discards temporal dynamics known to carry engagement information. The SCB engagement labels are derived from behaviour categories rather than direct human ratings, which may not fully align with teachers’ engagement judgements. Finally, results on 16 subjects in a webcam setting (DAiSEE) may not generalise to diverse classroom environments.

Understanding the gaps. The feature space analysis (Sec. 4.4) surfaces three fundamental gaps between current VLMs and individual engagement recognition. The *representation gap*: CLIP organises visual features around appearance-level cues—identity, lighting, background—because such cues dominate internet-scale image-caption training data, not engagement states. The *granularity gap*: the four-class engagement scale requires fine-grained ordinal discrimination that zero-shot VLMs are not designed for; the ≈ 0.07 cosine similarity range confirms that CLIP treats engagement levels as nearly synonymous in language space. The *temporal gap*: engagement is inherently dynamic, yet our single-frame protocol discards the most informative

temporal signal. Taken together, these gaps explain why near-random performance is not an artefact of prompt choice or dataset noise, but a structural consequence of the pre-training objective mismatch.

Future directions. These gaps suggest concrete research paths. *Supervised adaptation*: linear probing or lightweight adapter layers trained on even small labelled engagement datasets could reveal whether engagement signal exists in frozen CLIP features but is entangled with identity. *Contrastive disentanglement*: a projection that explicitly separates engagement from identity—positive pairs sharing engagement label across different students, negatives sharing identity across engagement levels—would directly address the clustering-by-identity failure mode we observe. *Temporal VLMs*: video-native models should be evaluated on full 10-second DAiSEE clips; our results motivate this as a priority, given the insufficient signal in static frames. *Scene-to-individual transfer*: since scene-level recognition is tractable ($\kappa \approx 0.60$), future work could explore top-down approaches that use scene-level estimates as a prior to refine individual-level inference. *Benchmark standardisation*: prompt sensitivity ($\Delta\kappa$) should be reported as an explicit evaluation axis in future engagement benchmarks, rather than a single-prompt result, to avoid over-optimistic reporting.

6. Conclusion

We presented a systematic zero-shot benchmark of four Vision-Language Models for classroom engagement recognition, evaluated across two datasets and three prompt designs. Our results establish that current VLMs cannot reliably classify individual student engagement—with Cohen’s κ never exceeding 0.10 on DAiSEE—but show substantially more promise at the scene level, where CLIP and GPT-4o reach $\kappa \approx 0.60$ on SCB. We characterise three failure modes (near-random ordinal agreement, class collapse, and extreme prompt sensitivity) and document a practical deployment barrier (GPT-4o safety refusals for individual face analysis). We hope these baselines serve as a rigorous starting point for the community, and that the identified failure modes motivate targeted improvements: scene-level pre-aggregation, prompt-robust fine-tuning, and temporal modelling to go beyond single-frame analysis.

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Supplementary Material

7. Full Prompt Texts

This supplementary provides the exact prompt text used for each model and dataset in our zero-shot benchmark, enabling full reproducibility.

7.1. SCB Dataset | GPT-4o, LLaVA-1.5, and Qwen2.5-VL-7B

These prompts were applied to the scene-level SCB images for GPT-4o, LLaVA-1.5, and Qwen2.5-VL-7B. Each image was submitted directly; the model returned a rating based on visible student behaviours.

P1 — Minimal

```
Given this image, rate the overall
classroom's engagement level based on
the
majority of students:

0 = not engaged
1 = barely engaged
2 = engaged
3 = highly engaged

Answer with just the number (0, 1, 2, or
3).
```

P2 — Rubric-Anchored

```
You are an educational observer. Assess
the overall classroom's engagement based
on the predominant student behaviors and
body language:

0 = Distracted and disinterested (e.g.,
using a phone, bowing the head)
1 = Passively present but not focused
(e.g., leaning over the table)
2 = Attentive and following along (e.g.,
reading, writing)
3 = Actively focused and alert (e.g.,
raising a hand)

Respond with only the number (0, 1, 2, or
3).
```

P3 — Chain-of-Thought

```
Analyze the students in this classroom
environment to assess their collective
learning engagement.
```

```
Step 1 (Behaviors): Describe the
predominant activities of the majority
of the
students in one detailed sentence. Focus
specifically on how they are interacting
with their environment (e.g., writing,
reading, resting heads, using devices,
raising hands).
```

```
Step 2 (Posture & Focus): Describe the
general body posture and visual attention
of the class in one detailed sentence
(e.g., sitting upright, leaning forward,
slumped over, gaze directed at desks or
away).
```

```
Step 3 (Rating): Based strictly on the
behaviors and posture described in Steps 1
and 2, rate the overall engagement level
of the majority of students:

0 = not engaged at all
1 = barely engaged
2 = engaged
3 = highly engaged
```

```
Format your response strictly as:
Behaviors: [Your sentence]
Posture: [Your sentence]
Rating: [0, 1, 2, or 3]
```

7.2. SCB Dataset | CLIP Text Templates

For CLIP (ViT-B/32) on SCB, classification uses cosine similarity between the scene image embedding and per-class text template embeddings.

7.3. DAiSEE Dataset | GPT-4o, LLaVA-1.5, and Qwen2.5-VL-7B

Analogous prompts were used for the individual-student DAiSEE frames. The wording was adapted from scene-level to individual-level assessment.

P1 — Minimal

Table 6. CLIP text templates for SCB scene-level engagement classification.

Prompt	Level	Text Template
P1	L0	<i>a classroom where the majority of students are not engaged at all</i>
	L1	<i>a classroom where the majority of students are barely engaged</i>
	L2	<i>a classroom where the majority of students are engaged</i>
	L3	<i>a classroom where the majority of students are highly engaged</i>
P2	L0	<i>a classroom where students are distracted, bowing heads, using phones, or completely disengaged</i>
	L1	<i>a classroom where students are passively present but not focused, leaning heavily on desks</i>
	L2	<i>a classroom where students are attentive, reading, writing, or following along</i>
	L3	<i>a classroom where students are actively focused, raising hands, showing curiosity</i>
P3	L0	<i>a classroom with bored, disinterested students showing disengaged posture</i>
	L1	<i>a classroom with indifferent students showing neutral, passive posture</i>
	L2	<i>a classroom with attentive, interested students showing focused posture</i>
	L3	<i>a classroom with excited, curious students showing highly alert posture</i>

Given this image of a student, rate their engagement level:
0 = not engaged
1 = barely engaged
2 = engaged
3 = highly engaged
Answer with just the number (0, 1, 2, or 3).

P2 — Rubric-Anchored

You are an educational observer. Assess this student's engagement based on their visible behavior and body language:
0 = Distracted and disinterested (e.g., looking away from the screen, yawning)
1 = Passively present but not focused (e.g., occasionally glancing at the screen)
2 = Attentive and following along (e.g., watching the screen, maintaining eye contact)
3 = Actively focused and alert (e.g., leaning forward, showing curiosity)
Respond with only the number (0, 1, 2, or 3).

P3 — Chain-of-Thought

Analyze the student in this image to assess their learning engagement.
Step 1 (Behaviors): Describe what the student appears to be doing in one sentence.
Step 2 (Posture & Focus): Describe the student's body posture and where their gaze appears directed in one sentence.
Step 3 (Rating): Based on the above, rate the student's engagement level:
0 = not engaged at all
1 = barely engaged
2 = engaged
3 = highly engaged
Format your response strictly as:
Behaviors: [Your sentence]
Posture: [Your sentence]
Rating: [0, 1, 2, or 3]

7.4. DAiSEE Dataset | CLIP Text Templates

For CLIP (ViT-B/32) on DAiSEE, classification uses cosine similarity between the individual frame embedding and per-class text template embeddings. The class whose template yields the highest similarity is the prediction.

Table 7. CLIP text templates for DAiSEE engagement classification.

Prompt	Level	Text Template
P1	L0	<i>a photo of a student who is not engaged at all, completely distracted</i>
	L1	<i>a photo of a student who is barely engaged, passively present</i>
	L2	<i>a photo of a student who is engaged and attentive</i>
	L3	<i>a photo of a student who is highly engaged, very focused and alert</i>
P2	L0	<i>a student looking away from the screen, yawning or distracted, showing no interest</i>
	L1	<i>a student passively sitting, occasionally glancing at the screen, low attention</i>
	L2	<i>a student watching the screen attentively, maintaining eye contact with the content</i>
	L3	<i>a student leaning forward, deeply focused, actively concentrating on the content</i>
P3	L0	<i>a bored and disinterested student with a blank or sleepy expression</i>
	L1	<i>a slightly disengaged student with a neutral, unfocused expression</i>
	L2	<i>an attentive student with an interested and alert facial expression</i>
	L3	<i>a highly focused student showing curiosity and active concentration</i>

7.5. BLIP-VQA Question Strings

BLIP-VQA receives a question string alongside the image. The free-text answer is mapped to an engagement level via a hand-crafted lookup table.

P1 — Minimal

How engaged is this student? Answer: not engaged, barely engaged, engaged, or highly engaged.

P2 — Rubric-Anchored

Based on the student's visible behavior, are they: (0) distracted and disinterested, (1) passively present, (2) attentive and following along, or (3) actively focused? Answer with 0, 1, 2, or 3.

P3 — Chain-of-Thought

Describe the student's facial expression and body posture, then rate their engagement as: not engaged, barely engaged, engaged, or highly engaged.